



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SREE MURUGAN AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

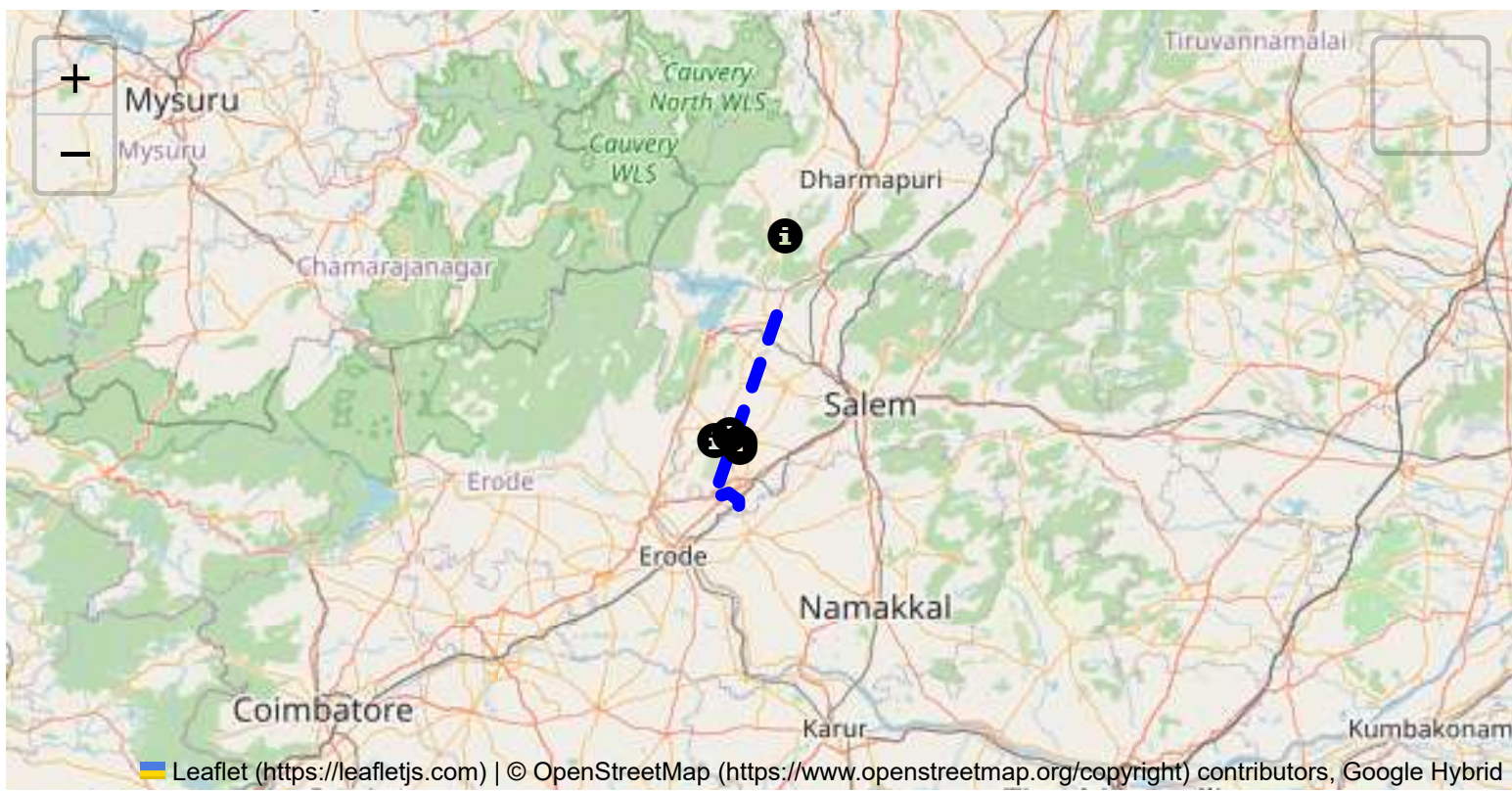
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

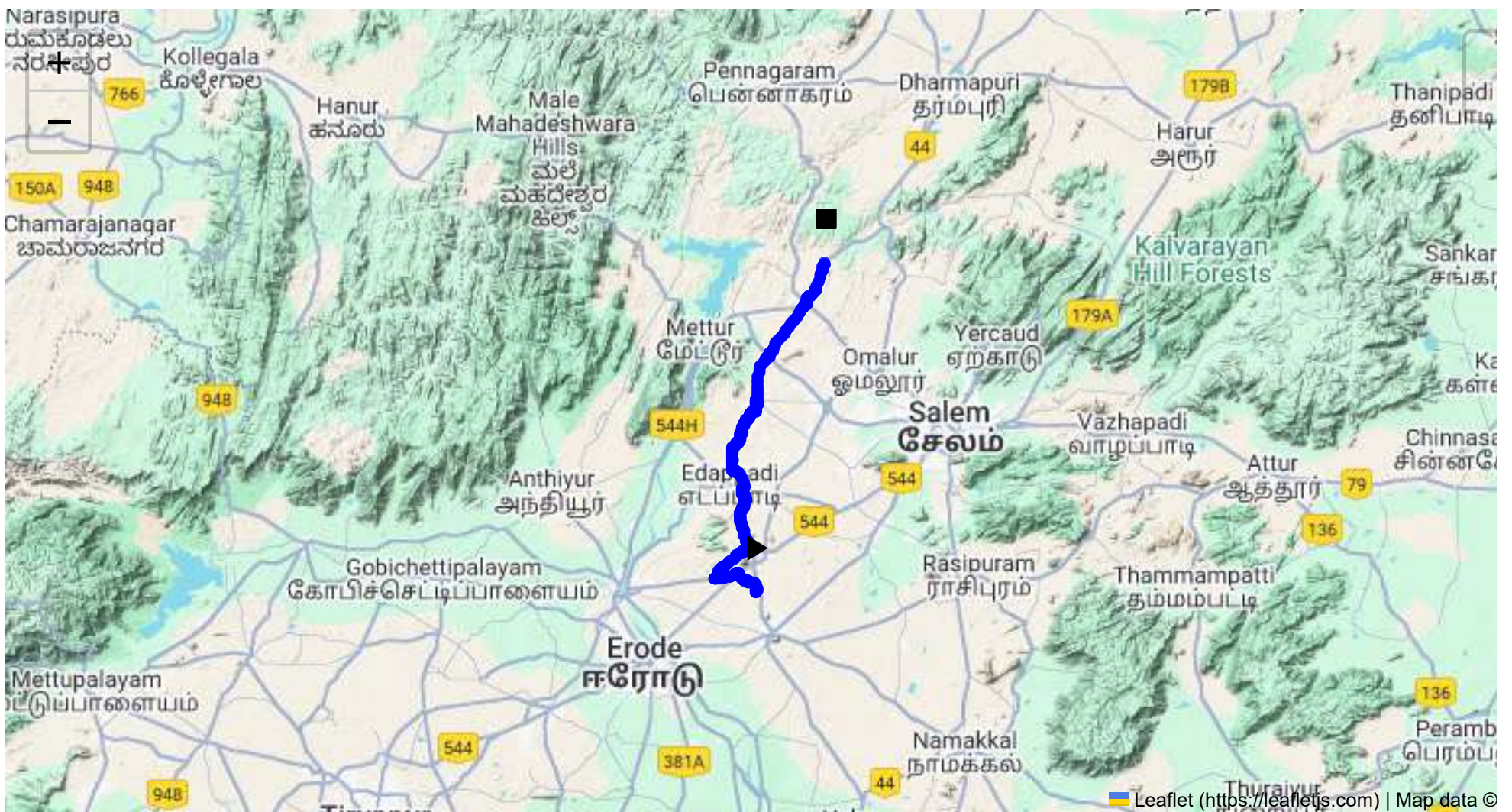
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 67.29 km
Estimated Duration: 1.6 hours
Adjusted Duration (Heavy Vehicle): 2.0 hours
Start: (11.4381, 77.8734)
End: (11.888036, 77.969828)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Vellar via the specified checkpoints is approximately 67.29 kilometers. It travels primarily through state highways and main roads, crossing through small towns and rural areas in Tamil Nadu. Key waypoints include the Sankari RS - Erode Privu Road, and Bhavani Main Road, stretching through agricultural regions and smaller settlements.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu typically experiences a tropical climate with varying weather patterns:

- **Summer (March to June):** High temperatures can reach up to 40°C. Heat can cause discomfort and mechanical strain on the vehicle.
- **Monsoon (July to September):** Heavy rains and potential flooding are common, increasing the risk of hydroplaning and road washouts.
- **Winter (October to February):** Mild temperatures, but early morning fog can reduce visibility, particularly in rural stretches.

3. Analysis of Traffic Patterns

- **Peak Hours:** Mornings (7 AM - 10 AM) and evenings (5 PM - 8 PM) see increased traffic due to local commuters.
- **Congestion-Prone Areas:** Smaller town zones, particularly near marketplaces and schools, may experience congestion. Bhavani Main Road might see higher volumes due to its connectivity.

4. Assessment of Road Quality and Infrastructure

- **General Assessment:** State highways and main roads in Tamil Nadu are largely well-maintained. However, rural stretches might have narrower lanes and lesser quality pavement that can affect heavy vehicle navigation.
- **Infrastructural Gaps:** Bridges and culverts may not always accommodate large trucks smoothly, especially after heavy rain.

5. Suggestions for Alternative Routes for Emergencies

- In emergencies, divert to NH44, a major north-south highway connecting to broader network routes offering faster and more reliable paths in urgent situations.
- Local roads via small towns like Tiruchengode can provide additional detours.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Regulations:** Transport of hazardous materials is guided by the Motor Vehicles Act and specific state regulations, including timing restrictions, required permits, and safety inspections.
- **Checks and Clearances:** Ensure valid permits are in hand and inspections are completed, especially when crossing district borders.

7. Overview of Historical Incidents

- Historical records show few major incidents, thanks to stringent regulations. Minor road spills or vehicle breakdowns are more common, largely managed locally without broader impacts.

8. Environmental Considerations and Sensitive Areas

- Farmlands and Waterbodies:** Nearby agricultural lands can be sensitive to chemical spills.
- Wildlife Considerations:** Navigate with caution near forest peripheries to avoid disrupting local fauna.

9. Analysis of Communication Coverage

- Coverage:** Generally strong in town areas, but could be spotty in rural sections especially between dense vegetation or hilly terrains.
- Advice:** Equipped with alternate communication devices (radios/satellite phones) in case of network failures.

10. Estimated Emergency Response Times

- Urban Areas:** 30-45 minutes due to proximity to services.
- Rural Areas:** May range from 45 minutes to 1 hour due to road conditions and remoteness.

11. Overall Summary of Risk Assessment

The route through Tamil Nadu from Sangagiri to Vellar for heavy vehicles carrying hazardous materials is generally manageable but demands caution, particularly considering potential weather disturbances, spotty communication in rural stretches, and ensuring compliance with local regulations. Alternative routes are accessible, and emergency response frameworks are active, albeit with variations in response times based on locality. Drivers should exercise vigilance, particularly in inclement weather and during peak traffic hours, to mitigate risks. Proper planning and navigation are crucial for safe transit in this region.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.60348, 77.83967	15 KM/Hr
2	Turn	Medium	11.43822, 77.87348	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	Medium	11.44900, 77.87406	30 KM/Hr
6	Turn	Medium	11.45357, 77.85789	30 KM/Hr
7	Turn	High	11.45352, 77.85698	15 KM/Hr
8	Turn	Medium	11.45847, 77.85140	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
9	Turn	Medium	11.45898, 77.85135	30 KM/Hr
10	Turn	Medium	11.46303, 77.84945	30 KM/Hr
11	Turn	Medium	11.46314, 77.84928	30 KM/Hr
12	Turn	High	11.45542, 77.81725	15 KM/Hr
13	Turn	High	11.45634, 77.81673	15 KM/Hr
14	Blind Spot	Blind Spot	11.49397, 77.86740	10 KM/Hr
15	Turn	Medium	11.49404, 77.86724	30 KM/Hr
16	Turn	Medium	11.50866, 77.86053	30 KM/Hr
17	Turn	Medium	11.52587, 77.85524	30 KM/Hr
18	Turn	Medium	11.54750, 77.85273	30 KM/Hr
19	Turn	Medium	11.55039, 77.85511	30 KM/Hr
20	Turn	Medium	11.55481, 77.85639	30 KM/Hr
21	Turn	High	11.57893, 77.85430	15 KM/Hr
22	Turn	Medium	11.57903, 77.85441	30 KM/Hr
23	Turn	Medium	11.58022, 77.85744	30 KM/Hr
24	Turn	Medium	11.60727, 77.84032	30 KM/Hr
25	Turn	Medium	11.60972, 77.84164	30 KM/Hr
26	Turn	Medium	11.64064, 77.84611	30 KM/Hr
27	Turn	Medium	11.64535, 77.85122	30 KM/Hr
28	Turn	Medium	11.65473, 77.85327	30 KM/Hr
29	Turn	Medium	11.66621, 77.85610	30 KM/Hr
30	Turn	Medium	11.67301, 77.86024	30 KM/Hr
31	Turn	Medium	11.67326, 77.86058	30 KM/Hr
32	Turn	Medium	11.67373, 77.86230	30 KM/Hr
33	Turn	Medium	11.68171, 77.86676	30 KM/Hr
34	Turn	High	11.68490, 77.86893	15 KM/Hr
35	Turn	High	11.68472, 77.87281	15 KM/Hr
36	Turn	High	11.69458, 77.87589	15 KM/Hr
37	Turn	Medium	11.69475, 77.87583	30 KM/Hr
38	Turn	High	11.69494, 77.87544	15 KM/Hr
39	Turn	High	11.69501, 77.87546	15 KM/Hr
40	Turn	High	11.69553, 77.87440	15 KM/Hr
41	Turn	Medium	11.69915, 77.87478	30 KM/Hr
42	Turn	Medium	11.69941, 77.87503	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
43	Turn	Medium	11.76013, 77.89028	30 KM/Hr
44	Turn	Medium	11.76020, 77.89070	30 KM/Hr
45	Blind Spot	Blind Spot	11.75947, 77.89214	10 KM/Hr
46	Turn	High	11.83365, 77.94484	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Parusuraman Hospital	11.697164, 77.877792	30 km/h	Medium
2	hospital	JP Nature Cure Hospital Siddha and Naturopathy	11.7002949, 77.8770897	30 km/h	Medium
3	hospital	SSS Hospital	11.7501471, 77.8832809	30 km/h	Medium
4	clinic	DSR Clinic	11.7603176, 77.8908853	30 km/h	Medium
5	hospital	Srilakshmi Hospital	11.7601, 77.892272	30 km/h	Medium
6	hospital	Government Hospital	11.758158, 77.894528	30 km/h	Medium
8	hospital	Vasu Speciality Hospital	11.8331097, 77.943169	30 km/h	Medium
9	hospital	Rajamanickam Hospital	11.8332408, 77.9431081	30 km/h	Medium
10	hospital	Child Hospital	11.833001, 77.943415	30 km/h	Medium
11	hospital	Naveen Nursing Home	11.8335355, 77.9437441	30 km/h	Medium
12	clinic	Sri Krishna Clinic	11.835443, 77.9450587	30 km/h	Medium
13	hospital	Sri Sakthi Hospital	11.835714, 77.9451014	30 km/h	Medium
14	hospital	Selvakumar Hospital	11.836215, 77.945842	30 km/h	Medium
15	clinic	SK Clinic	11.83619, 77.9457752	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	college	Kavery Educational Institutions	11.8255319, 77.9342256	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58300, 77.85813



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.60348, 77.83967



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45357, 77.85789



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45898, 77.85135



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46314, 77.84928



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45634, 77.81673



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.49397, 77.86740



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49404, 77.86724



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.50866, 77.86053



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.52587, 77.85524



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.54750, 77.85273



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55039, 77.85511



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55481, 77.85639



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.57893, 77.85430



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57903, 77.85441



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58022, 77.85744



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60727, 77.84032



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60972, 77.84164



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64064, 77.84611



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64535, 77.85122



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65473, 77.85327



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66621, 77.85610



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67301, 77.86024



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67326, 77.86058



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67373, 77.86230



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68171, 77.86676



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.68490, 77.86893



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.68472, 77.87281



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69458, 77.87589



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69475, 77.87583



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69494, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69501, 77.87546



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69553, 77.87440



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69915, 77.87478



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69941, 77.87503



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.76013, 77.89028



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.76020, 77.89070



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.75947, 77.89214



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.83365, 77.94484
